

5-2c Comparative Advantage

According to Adam Smith, each nation should look for absolute advantage. However, what can nations do when they do *not* possess absolute advantage? Continuing our two-country example of China and the United States, what if China is absolutely inferior to the United States in the production of both wheat and aircraft (which is the real case today)? What should China do? What should the United States do? Obviously, the theory of absolute advantage runs into a dead end.

In response, British economist David Ricardo developed a theory of comparative advantage (A theory that focuses on the relative (not absolute) advantage in one economic activity that one nation enjoys in comparison with other nations.)

in 1817. This theory suggests that even though the United States has an absolute advantage over China in both wheat and aircraft, as long as China is not equally less efficient in the production of both goods, China can still choose to specialize in the production of one good (such as wheat) where it has

comparative advantage (Relative (not absolute) advantage in one economic activity that one nation enjoys in comparison with other nations.)

—defined as the relative (not absolute) advantage in one economic activity that one nation enjoys in comparison with other nations. [Figure 5.3](#) and [Table 5.3](#) show that China's comparative advantage lies in its *relatively less inefficient* production of wheat: If China devotes all resources to wheat, it can produce 10,000 tons, which is four-fifths of the 12,500 tons the United States can produce. However, at a maximum, China can only produce 20 aircraft, which is merely half of the 40 aircraft the United States can make. By letting China specialize in the production of wheat and importing some wheat from China, the United States is able to leverage its strengths by devoting its resources to aircraft. For example, if

- (1) the United States devotes four-fifths of its resources to aircraft and one-fifth to wheat (Point C in [Figure 5.3](#)),
- (2) China concentrates 100% of its resources on wheat (Point E), and
- (3) they trade with each other, both countries produce and consume more than what they would produce and consume if they inefficiently devote half of their resources to each activity.

Figure 5.3

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Table 5.3

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**Total units of resources = 800
for each country**

		Wheat	Aircraft
(1) Resources required to produce 1,000 tons of wheat and 1 aircraft.	China	80	40
		resources	resources
	US	64	20
		resources	resources
(2) Production and consumption with no specialization and without trade (each country devotes <i>half</i> of its resources to each activity).	China (point F)	5,000 tons	10 aircraft
		6,250 tons	20 aircraft
	US (point B)	11,250 tons	30 aircraft
	<i>Total production</i>		
(3) Production with specialization (China devotes all resources to wheat, and the United States devotes one-fifth of its resources to wheat and four-fifths of its resources to aircraft)	China (point E)	10,000 tons	0
			32 aircraft
	US (point C)	2,500 tons	32 aircraft
	<i>Total production</i>	12,500 tons	

Total units of resources = 800

for each country

Wheat

Aircraft

(4) Consumption after China trades 4,000 tons of wheat for 11 US aircraft while producing at points E and C, respectively (Scenario #3).	China	6,000 tons	11 aircraft
	US	6,500 tons	21 aircraft
	<i>Total consumption</i>	12,500 tons	32 aircraft

(5) <i>Gains</i> from trade: Increase in consumption as a result of specialization and trade (Scenario #4 versus #2).	China	11,000 tons	11 aircraft
	US	1250 tons	11 aircraft

Again, there are *net* gains from trade, this time from comparative advantage. One crucial concept here is

opportunity cost (Cost of pursuing one activity at the expense of another activity, given the alternatives (other opportunities).)

—given the alternatives (other opportunities), the cost of pursuing one activity at the expense of another activity. For the United States, the opportunity cost of concentrating on wheat at point A in [Figure 5.3](#) is tremendous relative to producing aircraft at point D, because it is only 25% more productive in wheat than China, but is 100% more productive in aircraft.

Relative to absolute advantage, the theory of comparative advantage seems counterintuitive. However, this theory is far more realistic and useful in the real world. This is because, while it is easy to identify an absolute advantage in a highly simplified, two-country world as in [Figure 5.2](#), how can each nation decide what to specialize in when there are more than 200 nations in the world? It is simply too challenging to ascertain that one nation is absolutely better than all others in one activity. Is the United States *absolutely* better than not only China, but also all other 200 nations in aircraft production? European nations that produce Airbus obviously beg to differ. The theory of comparative advantage suggests that even without an absolute advantage, the United States can still profitably specialize in aircraft as long as it is relatively more efficient than others. This insight thus has greatly lowered the threshold for specialization.

Where do absolute and comparative advantages come from? In one word, productivity. Smith looked at *absolute* productivity differences, and Ricardo emphasized *relative* productivity differences—in this sense, absolute advantage is really a special case of comparative advantage. But what leads to such productivity differences? In the early 20th century, Swedish economists Eli Heckscher and Bertil Ohlin argued that absolute and comparative advantages stem from different

factor endowments (The extent to which different countries possess various factors of production such as labor, land, and technology.)

, namely, the extent to which different countries possess various factors, such as labor, land, and technology. This

factor endowment theory (or Heckscher-Ohlin theory) (A theory that suggests that nations will develop comparative advantages based on their locally abundant factors.)

proposed that nations will develop comparative advantage based on their *locally abundant* factors. Numerous examples support the theories of comparative advantage and factor endowments. For example, Brazil is blessed by an abundant supply of sunshine, soil, and water, which make it a world-class player in agricultural products. When Nissan makes cars in Japan, it uses a lot of robots. But Nissan in India relies more heavily on labor to make cars. In another example, when Indian firms set up call centers to service Western clients, they use human labor, a factor that is abundant in India, to replace some automation functions when answering the phone—telephone automation technology has been developed in the West because of labor shortage. Western clients are happier, because they can actually talk to a live person instead of talking to or dialing to a machine (dial 1 for this, dial 2 for that).

In summary, *classical* theories—

- (1) mercantilism,
- (2) absolute advantage, and
- (3) comparative advantage (which includes factor endowments)—had evolved from approximately 300 to 400 years ago to the beginning of the 20th century.

More recently, three *modern* theories, outlined next, emerged.